

3 June 2026

Dr Guillaume Martin
Editor, *GENETICS*

Re: Revised manuscript GENETICS-2026-309068, “Additive Channels in Curved Fitness Landscapes”

Dear Dr Martin,

Thank you for the opportunity to revise our manuscript, “Additive Channels in Curved Fitness Landscapes,” by Daniel Ortiz-Barrientos and Mark Cooper. We are pleased to submit the revised manuscript, together with a detailed response to the editor and reviewers.

We are grateful for the careful and constructive feedback. In response, we have revised the manuscript substantially to clarify the scope of the work and to make the connection to quantitative-genetic theory more explicit. The revised version now frames the problem as a breeding-value-to-fitness question, rather than as a claim about the mechanistic genotype-to-phenotype map. We have also strengthened the abstract, introduction, notation, and the relationship to Lande-Arnold selection gradients, Fisher’s average effects, and related geometric treatments.

The manuscript develops the additivity index, A_g , as a local diagnostic for when linear additive prediction is expected to work on a curved fitness surface. It defines additive channels as regions of breeding-value space in which local curvature contributes relatively little to heritable log-fitness variation, and it uses a coupled selection-inheritance framework to identify conditions under which populations enter, persist in, or leave such regimes.

We have also prepared a focused reproducibility repository containing the retained figure-generation code, final figure files, manifest files, and a runner that verifies the expected outputs: <https://github.com/dortizbarrientos/additive-channels-fitness-landscapes>

We confirm that the revised manuscript is original, is not under consideration elsewhere, and that both authors have approved this resubmission.

Thank you again for considering our revised manuscript.

Sincerely,

Daniel Ortiz-Barrientos

Professor in Evolutionary Genetics
School of the Environment
The University of Queensland

on behalf of Daniel Ortiz-Barrientos and Mark Cooper